Final Exam Practice Session

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Part I: Complex Numbers and Analytic Functions

Exercise 1. Find all complex roots of the equation $z^4 = -16$. Express your answers in exponential form and sketch them in the complex plane.

Exercise 2. Evaluate the complex limit: $\lim_{z \to 2i} \dfrac{z^2+4}{z-2i}$.

Exercise 3. Consider the function $f(z) = x^3 + iy^3$. Using the Cauchy-Riemann equations, determine all points where $f(z)$ is differentiable, and determine all domains where $f(z)$ is analytic.

Exercise 4. Verify that $u(x, y) = e^{-y}\sin(x)$ is a harmonic function. Then, find its harmonic conjugate $v(x, y)$ such that $f(z) = u + iv$ is analytic.

Part II: Elementary Functions and Complex Integration

Exercise 5. Using the principal branch of the complex logarithm, compute the principal value of $(-1+i)^i$. Express your answer in the form $x + iy$.

Exercise 6. Determine and sketch the branch cut for the function $f(z) = \operatorname{Log}(z + 2 - i)$.

Exercise 7. Evaluate the contour integral $\int_C \bar{z}\, dz$, where C is the parabolic arc $y = x^2$ from $z = 0$ to $z = 1 + i$.

Exercise 8. Use the generalized Cauchy Integral Formula to evaluate $\displaystyle\oint_{|z|=2} \frac{e^{iz}}{(z-\pi/2)^2}\, dz$, traversed counterclockwise.

Part III: Series and Singularities

Exercise 9. Find the Laurent series expansion for $f(z) = \frac{1}{(z-1)(z-3)}$ valid in the annulus $1 < |z| < 3$.

Exercise 10. Classify the singularity at $z = 0$ for the function $f(z) = z^2 e^{1/z}$. Is it removable, a pole, or essential? Compute its residue.

Exercise 11. Identify all singularities and compute the residue at each for $f(z) = \frac{z^2}{(z-i)^2(z+2)}$.

Exercise 12. Use the Residue at Infinity to evaluate $\displaystyle\oint_{|z|=4} \frac{1}{z^2(z+1)}\, dz$.

Part IV: Real Integrals and Conformal Mapping

Exercise 13. Use the Residue Theorem to evaluate the definite trigonometric integral: $\displaystyle\int_0^{2\pi} \frac{1}{5 + 3\cos\theta}\, d\theta$.

Exercise 14. Evaluate the improper real integral: $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^2+2x+2}\, dx$.

Exercise 15. Evaluate the Fourier-type integral: $\displaystyle\int_{-\infty}^{\infty} \frac{\cos(x)}{x^2+1}\, dx$.

Exercise 16. Find the unique Möbius transformation $w = T(z)$ that maps $z_1 = 0 \mapsto w_1 = -1$, $z_2 = i \mapsto w_2 = 0$, and $z_3 = -i \mapsto w_3 = \infty$.

Exercise 17. Show that the mapping $w = z^2$ is conformal everywhere except at $z = 0$.

Exercise 18. Find a bounded harmonic function $h(x, y)$ in the upper half-plane $(y > 0)$ that satisfies the boundary conditions $h(x, 0) = 0$ for $x < 0$ and $h(x, 0) = 10$ for $x > 0$.